

Week 2 Module

1* Fraud Detection (Linear Search)

```
import java.util.Scanner;

public class FraudDetection {

    public static void main (String[] args) {

        Scanner sc = new Scanner (System.in);

        int n = sc.nextInt();

        int [] transactions = new int [n];

        for (int i = 0; i < n; i++) {

            transactions [i] = sc.nextInt();

        }

        int suspiciousID = sc.nextInt();

        boolean found = false;

        for (int i = 0; i < n; i++) {

            if (transactions [i] == suspiciousID) {

                found = true;

                break;

            }

        }

        if (found) {

            System.out.println ("Fraud Transaction");

        }

        else

            System.out.println ("Transaction not Found");

    }

}
```



Welcome

J FraudTransactionDetection.java X



J FraudTransactionDetection.java > ...

```
1  import java.util.Scanner;
2
3  public class FraudTransactionDetection {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7
8          // Input: number of transactions
9          int n = sc.nextInt();
10         int[] transactions = new int[n];
11
12         // Input: transaction IDs
13         for (int i = 0; i < n; i++) {
14             transactions[i] = sc.nextInt();
15         }
16
17         // Input: suspicious transaction ID
18         int suspiciousID = sc.nextInt();
19
20         // Linear search
21         boolean found = false;
22         for (int i = 0; i < n; i++) {
23             if (transactions[i] == suspiciousID) {
24                 found = true;
25                 break;
26             }
27         }
28
29         // Output
30         if (found)
31             System.out.println("Fraud Transaction Found");
32         else
33             System.out.println("Transaction Not Found");
34         sc.close();
35     }
36 }
37
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\MISTHI\OneDrive\Desktop\Week2Module.java> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-X
paceStorage\fb413c53588f222a40096bb76998e25c\redhat.java\jdt_ws\Week2Module.java_372e3352\bin' 'FraudTransac
1201 1305 1450 1408 1502
1450
```

Fraud Transaction Found

PS C:\Users\MISTHI\OneDrive\Desktop\Week2Module.java> []



2. Server log Search (Binary Search)

```
import java.util.Scanner;
public class ServerLogSearch {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);
        int n = sc.nextInt();
        int[] logs = new int [n];
        for (int i=0; i<n; i++) {
            logs[i] = sc.nextInt();
        }
        int target = sc.nextInt();
        int low = 0; high = n-1;
        boolean found = false;
        while (low <= high) {
            int mid = (low + (high - low) / 2);
            if (logs[mid] == target) {
                found = true;
                break;
            } else if (logs[mid] < target) {
                low = mid + 1;
            } else {
                high = mid - 1;
            }
        }
        if (found) { SOP ("Log Found"); }
        else { SOP ("Log Not Found"); }
    }
}
```



Welcome

FraudTransactionDetection.java

ServerLogSearch.java X



ServerLogSearch.java > ServerLogSearch > main(String[])

```
1  import java.util.Scanner;
2
3  public class ServerLogSearch {
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6
7          // Input: number of log IDs
8          int n = sc.nextInt();
9          int[] logs = new int[n];
10
11         // Input: sorted log IDs
12         for (int i = 0; i < n; i++) {
13             logs[i] = sc.nextInt();
14         }
15
16         // Input: target log ID
17         int target = sc.nextInt();
18
19         // Binary Search
20         int low = 0, high = n - 1;
21         boolean found = false;
22
23         while (low <= high) {
24             int mid = low + (high - low) / 2;
25             if (logs[mid] == target) {
26                 found = true;
27                 break;
28             } else if (logs[mid] < target) {
29                 low = mid + 1;
30             } else {
31                 high = mid - 1;
32             }
33         }
34
35         // Output
36         if (found)
37             System.out.println("Log Found");
38         else
39             System.out.println("Log Not Found");
40
41         sc.close();
42     }
43 }
44
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\WISTHI\OneDrive\Desktop\Week2Module.java> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' ^-XX:+ShowCodeDetailsInException'
```

7

```
1001 1005 1010 1015 1020 1030 1040
```

```
1015
```

```
Log Found
```

```
PS C:\Users\WISTHI\OneDrive\Desktop\Week2Module.java> |
```


36 Sort Employee Salaries (Bubble Sort)

```
import java.util. Scanner ;
public class SortSalaries {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);
        int n = sc.nextInt();
        int [] salaries = new int [n];
        for (int i=0; i<n; i++) {
            salaries [i] = sc.nextInt();
        }
        for (int i=0; i<n-1; i++) {
            for (int j=0; j<n-i-1; j++) {
                if (salaries [j] > salaries [j+1]) {
                    int temp = salaries [j];
                    salaries [j] = salaries [j+1];
                    salaries [j+1] = temp;
                }
            }
        }
        for (int i=0; i<n; i++) {
            System.out.print (salaries [i] + " ");
        }
        sc.close ();
    }
}
```



Welcome

J FraudTransactionDetection.java

J ServerLogSearch.java

J SortSalaries.java X



J SortSalaries.java > ...

```
1  import java.util.Scanner;
2
3  public class SortSalaries {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7
8          // Input: number of employees
9          int n = sc.nextInt();
10         int[] salaries = new int[n];
11
12         // Input: salaries
13         for (int i = 0; i < n; i++) {
14             salaries[i] = sc.nextInt();
15         }
16
17         // Bubble Sort
18         for (int i = 0; i < n - 1; i++) {
19             for (int j = 0; j < n - i - 1; j++) {
20                 if (salaries[j] > salaries[j + 1]) {
21                     // swap
22                     int temp = salaries[j];
23                     salaries[j] = salaries[j + 1];
24                     salaries[j + 1] = temp;
25                 }
26             }
27         }
28
29         // Output: sorted salaries
30         for (int i = 0; i < n; i++) {
31             System.out.print(salaries[i] + " ");
32         }
33
34         sc.close();
35     }
36 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\MISTHI\OneDrive\Desktop\Week2Module.java> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe'
8f222a40096bb76998e25c\redhat.java\jdt_ws\Week2Module.java_372e3352\bin' 'SortSalaries'
5
45000 32000 55000 28000 60000
28000 32000 45000 55000 60000
PS C:\Users\MISTHI\OneDrive\Desktop\Week2Module.java> 
```



4) Sort Website Response Time (Insertion Sort)

```
public class SortResponseTimes {  
    public static void main (String[] args) {  
        Scanner sc = new Scanner (System.in);  
        int n = sc.nextInt();  
        int[] times = new int [n];  
        for (int i = 0; i < n; i++) {  
            times[i] = sc.nextInt();  
        }  
        for (int i = 1; i < n; i++) {  
            int key = times[i];  
            int j = i - 1;  
            while (j >= 0 && times[j] > key) {  
                times[j+1] = times[j];  
                j--;  
            }  
            times[j+1] = key;  
        }  
        for (int i = 0; i < n; i++) {  
            System.out.print(times[i] + " ");  
        }  
        sc.close();  
    }  
}
```



J SortResponseTimes.java > ...

```
1  import java.util.Scanner;
2
3  public class SortResponseTimes {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7
8          // Input: number of response times
9          int n = sc.nextInt();
10         int[] times = new int[n];
11
12         // Input: response times
13         for (int i = 0; i < n; i++) {
14             times[i] = sc.nextInt();
15         }
16
17         // Insertion Sort
18         for (int i = 1; i < n; i++) {
19             int key = times[i];
20             int j = i - 1;
21
22             // Shift elements greater than key to the right
23             while (j >= 0 && times[j] > key) {
24                 times[j + 1] = times[j];
25                 j--;
26             }
27             times[j + 1] = key;
28         }
29
30         // Output: sorted response times
31         for (int i = 0; i < n; i++) {
32             System.out.print(times[i] + " ");
33         }
34
35         sc.close();
36     }
37 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\MISTHI\OneDrive\Desktop\Week2Module.java> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-X
8f222a40096bb76998e25c\redhat.java\jdt_ws\Week2Module.java_372e3352\bin' 'SortResponseTimes'
```

5

250 120 320 180 100

100 120 180 250 320

```
PS C:\Users\MISTHI\OneDrive\Desktop\Week2Module.java> 
```



5) Prioritize Bug Reports (Selection Sort)

```
import java.util.Scanner;
public class BugReportPriorities {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);
        int n = sc.nextInt();
        int [] priorities = new int [n];
        for (int i=0; i<n; i++) {
            priorities[i] = sc.nextInt();
        }
        for (int i=0; i<n-1; i++) {
            int minIndex = i;
            for (int j=i+1; j<n; j++) {
                if (priorities[j] < priorities[minIndex])
                    minIndex = j;
            }
            int temp = priorities [minIndex];
            priorities [minIndex] = priorities [i];
            priorities [i] = temp;
        }
        for (int i=0; i<n; i++) {
            System.out.print (priorities[i] + " ");
        }
    }
}
```



Welcome

J FraudTransactionDetection.java

J ServerLogSearch.java

J SortSalaries.java

J SortR



J BugReportPriorities.java > ...

```
1  import java.util.Scanner;
2
3  public class BugReportPriorities {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7
8          // Input: number of bug reports
9          int n = sc.nextInt();
10         int[] priorities = new int[n];
11
12         // Input: priority scores
13         for (int i = 0; i < n; i++) {
14             priorities[i] = sc.nextInt();
15         }
16
17         // Selection Sort
18         for (int i = 0; i < n - 1; i++) {
19             int minIndex = i;
20             for (int j = i + 1; j < n; j++) {
21                 if (priorities[j] < priorities[minIndex]) {
22                     minIndex = j;
23                 }
24             }
25             // Swap
26             int temp = priorities[minIndex];
27             priorities[minIndex] = priorities[i];
28             priorities[i] = temp;
29         }
30
31         // Output: sorted priorities
32         for (int i = 0; i < n; i++) {
33             System.out.print(priorities[i] + " ");
34         }
35
36         sc.close();
37     }
38 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\MISTHI\OneDrive\Desktop\Week2Module.java> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' -cp 'C:\Program Files\Java\jdk-25.0.2\bin\java\998e25c\redhat.java\jdt_ws\Week2Module.java_372e3352\bin' 'BugReportPriorities'
```

```
5
9 2 8 1 5
1 2 5 8 9
```

```
PS C:\Users\MISTHI\OneDrive\Desktop\Week2Module.java> 
```

